Module – 2: Conditional and Joint Entropy, and Mutual Information

(Joint entropy, Conditional entropy, Mutual information, Discrete memoryless channels - BSC, BEC, noise-free channel, Channel with independent I/O, Cascaded channels. Channel Capacity. Kullback–Leibler divergence (Relative Entropy), Cross-Entropy, Jensen–Shannon divergence)

2.1 Joint Probability $P(x, y)$: Probability that (x, y) simultaneously occur, where x and y represent the random variable

2.2 Conditional Probability $P(x/y)$: $P(x/y)$ = represent the conditional probability (Probability of x when y is known)

$$P(x, y) = P(x/y) \,.\, P(y)$$

or

$$P(x, y) = P(y/x) \,.\, P(x)$$

If x and y are independent, then $P(x, y) = P(x) \,.\, P(y)$

2.3 Joint Entropy $H(X, Y)$: The joint entropy of two random variables X and Y measures the total uncertainty associated with the pair (X, Y). Let X be the channel input with N symbols and Y be the channel output with M symbols then the joint Entropy is:

$$H(X,Y) = \sum_{i=1}^{N}\sum_{j=1}^{M} p(x_i, y_j)\log_2\left(\frac{1}{p(x_i, y_j)}\right) = -\sum_{i=1}^{N}\sum_{j=1}^{M} p(x_i, y_j)\log_2\left(p(x_i, y_j)\right)$$

- It measures how much information is needed to describe both the transmitted symbol (X) and the received symbol (Y).
- It captures both the uncertainty in the input source, and the additional uncertainty introduced by the channel noise.
- If X and Y are independent, then $H(X, Y) = H(X) + H(Y)$.

2.4 Conditional Entropy: Conditional entropy measures the average uncertainty remaining in a random variable after observing another variable.

2.4.1 Channel Context $H(Y/X)$: The conditional entropy of output Y for the known input X is:

$$H(Y/X) = -\sum_{i=1}^{N}\sum_{j=1}^{M} p(x_i, y_j)\log_2\left(p(y_j/x_i)\right)$$

- It measures the uncertainty in the received symbol Y when the transmitted symbol X is known.
- Represents the noise level of the communication channel.
- If the channel is noise-free, then $p(y/x) = 1 \rightarrow H(Y/X) = 0$.
- If the channel is very noisy, then $H(Y/X)$ increases.

2.4.2 Reverse Channel Uncertainty *H*(*X*/*Y*): The conditional entropy of input X for the known output Y is:

$$H(X/Y) = -\sum_{i=1}^{N}\sum_{j=1}^{M} p(x_i, y_j)\log_2\left(p(x_i / y_j)\right)$$

- Measures the uncertainty in what was transmitted after observing the received symbol.
- Represents the receiver's confusion due to channel noise.
- If the channel is perfect (If Y completely determines X), then $H(X|Y) = 0$.
- If the channel is noisy, then $H(X|Y) > 0$.
- If Y knows no information about X, then $H(X/Y) = H(X)$.
- Relationship between them:

$$H(X, Y) = H(Y) + H(X|Y)$$

or

$$H(X, Y) = H(X) + H(Y|X)$$

Chain rule

The joint entropy can be written as the sum $H(X, Y) = H(X) + H(Y|X)$

Proof:

$$H(X,Y) = -\sum_{i=1}^{N}\sum_{j=1}^{M} p(x_i, y_j)\log p(x_i, y_j)$$

$$H(X,Y) = -\sum_{i=1}^{N}\sum_{j=1}^{M} p(x_i, y_j)\log\left(p(x_i)p(y_j | x_i)\right)$$

$$H(X,Y) = -\sum_{i=1}^{N}\sum_{j=1}^{M} p(x_i, y_j)\log p(x_i) - \sum_{i=1}^{N}\sum_{j=1}^{M} p(x_i, y_j)\log p(y_j | x_i)$$

$$H(X,Y) = -\sum_{i=1}^{N} p(x_i)\log p(x_i) - \sum_{i=1}^{N}\sum_{j=1}^{M} p(x_i, y_j)\log p(y_j | x_i)$$

$H(X, Y) = H(X) + H(Y|X)$

Mutual Information: Mutual Information is a fundamental concept in information theory that measures how much information one random variable contains about another. Mutual Information between two events (or symbols) x_i and y_j is defined as:

$$I(x_i; y_j) = \log_2\left[\frac{p(x_i / y_j)}{p(x_i)}\right]$$

- It measures how much information the received symbol y_j provides about the transmitted symbol x_i.

- If knowing y_j greatly reduces uncertainty about x_i, then mutual information is high.
- If y_j gives no clue about x_i, then mutual information is zero.

2.5 Average Mutual Information: Mutual information for individual symbols is not enough; communication happens repeatedly. So, we compute the average mutual information over all possible symbol pairs (x_i, y_j):

$$I(X;Y) = \sum_{i=1}^{N}\sum_{j=1}^{M} p(x_i, y_j)\log_2\left[\frac{p(x_i / y_j)}{p(x_i)}\right]$$

- It is the average information that the received signal Y conveys about the transmitted signal X. It reflects the overall efficiency of the channel in delivering information.

2.6 Relationship to Entropy: Mutual information can also be expressed as:

$$I(X;\, Y) = H(X) - H(X/Y)$$

or

$$I(X;\, Y) = H(Y) - H(Y/X)$$

or

$$I(X;\, Y) = H(X) + H(Y) - H(X,\, Y)$$

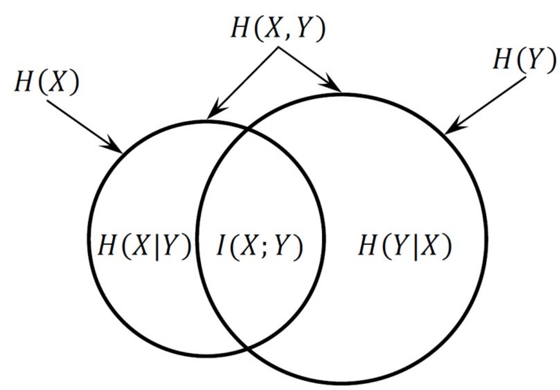


Fig: Relationship between mutual information and joint and conditional Entropies

2.7 Important Properties of Mutual Information:

Property	Details
Symmetry	$I(X;\, Y) = I(Y;\, X)$, Mutual information is symmetric. The information X gives about Y equals the information Y gives about X.
Non-negative	$I(X;\, Y) \geq 0$ Mutual information is always zero or positive. $I(X;\, Y) = 0$, only when X and Y are independent. Otherwise, it is strictly positive.
Relation with Entropy	$I(X;\, Y) = H(X) - H(X\|Y)$ $I(X;\, Y) = H(Y) - H(Y\|X)$ Mutual information = reduction in uncertainty of one variable due to the other.

Joint Entropy Identity	$I(X;Y) = H(X) + H(Y) - H(X,Y)$ Shows how mutual information connects with joint entropy.
Bounds of Mutual Information	$0 \le I(X;Y) \le \min[H(X), H(Y)]$ Mutual information cannot exceed the entropy of either variable.
Zero Mutual Information for Independent Variables	If X and Y are independent: $P(x, y) = P(x)\,P(y)$ $I(X;Y) = 0$ No information about one is obtained from the other.
Maximum Mutual Information in Noise-Free Channels	If Y is a perfect copy of X: $H(X\|Y) = 0$ $I(X;Y) = H(X)$ All information sent is received ideal communication.

2.8 Discrete Memory-less Channel: A Discrete Memoryless Channel (DMC) is one of the most fundamental models in digital communication. It describes how symbols transmitted through a channel may be altered due to noise, but without any dependence on past transmissions.

- A DMC is a communication channel where input and output alphabets are discrete: Input symbols: $X = \{x_1, x_2, \ldots, x_N\}$ and Output symbols $Y = \{y_1, y_2, \ldots, y_M\}$.
- The channel has no memory: The output at any time depends only on the current input symbol, not on previous inputs or outputs.
- Characterized by transition probabilities $P(y_j \mid x_i)$. This represents the probability that symbol y_j is received when x_i was transmitted.
- All transition probabilities can be grouped into a matrix called Channel Matrix (also called as Noise matrix or Channel transition matrix):

$$P(Y \mid X) = \begin{bmatrix} P(y_1 \mid x_1) & P(y_2 \mid x_1) & \cdots & P(y_M \mid x_1) \\ P(y_1 \mid x_2) & P(y_2 \mid x_2) & \cdots & P(y_M \mid x_2) \\ \vdots & \vdots & \ddots & \vdots \\ P(y_1 \mid x_N) & P(y_2 \mid x_N) & \cdots & P(y_M \mid x_N) \end{bmatrix}$$

- In the channel transition matrix, sum of all elements in each row is equal to ‘1’.

$$\sum_{j=1}^{M} P(y_j \mid x_i) = 1$$

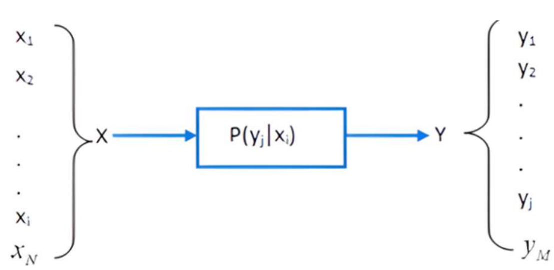


Fig. Discrete Memoryless Channel.

2.9 Joint Probability Matrix

A Joint Probability Matrix represents the joint distribution of two discrete random variables X and Y. It shows the probability of each possible pair (x_i, y_j) occurring simultaneously. If $X = \{x_1, x_2, \ldots, x_N\}$ be the input alphabet and $Y = \{y_1, y_2, \ldots, y_M\}$ be the output alphabet, then the joint probability matrix is:

$$P(X,Y) = \begin{bmatrix} P(x_1,y_1) & P(x_1,y_2) & \cdots & P(x_1,y_M) \\ P(x_2,y_1) & P(x_2,y_2) & \cdots & P(x_2,y_M) \\ \vdots & \vdots & \ddots & \vdots \\ P(x_N,y_1) & P(x_N,y_2) & \cdots & P(x_N,y_M) \end{bmatrix}$$

Each entry satisfies:

$$P(x_i,y_j) \geq 0, \sum_{i=1}^{N}\sum_{j=1}^{M} P(x_i,y_j) = 1$$

The sum of all elements in the i^{th} row gives the probability of the i^{th} input symbol.

$$P(x_i) = \sum_{j=1}^{M} P(x_i,y_j)$$

The sum of all elements in the j^{th} column gives the probability of the j^{th} output symbol.

$$P(y_j) = \sum_{i=1}^{N} P(x_i,y_j)$$

The relationship between conditional probability and joint probabilities is as follows:

$$P(y_j \mid x_i) = \frac{P(x_i,y_j)}{P(x_i)}$$

For the channel matrix given below compute the conditional Entropy $H(Y \mid X)$:

$$P(y \mid x) = \begin{bmatrix} 0.5 & 0.3 & 0.2 \\ 0.1 & 0.4 & 0.5 \end{bmatrix}$$

If the input symbol probabilities are $P(x_1) = 0.6, P(x_2) = 0.4$.

Ans. Step 1: Compute row entropies

For x_1:

$$H_1 = -[0.5\log_2 0.5 + 0.3\log_2 0.3 + 0.2\log_2 0.2]$$
$$H_1 = 0.5 + 0.521 + 0.464 = 1.485$$

For x_2:

$$H_2 = -[0.1\log_2 0.1 + 0.4\log_2 0.4 + 0.5\log_2 0.5]$$
$$H_2 = 0.332 + 0.529 + 0.5 = 1.361$$

Step 2: Compute total conditional entropy

$$H(Y/X) = -\sum_{i=1}^{N}\sum_{j=1}^{M} p(x_i,y_j)\log_2\left(p(y_j/x_i)\right)$$

$$H(Y \mid X) = -\sum_{i=1}^{N} p(x_i) \sum_{j=1}^{M} p(y_j \mid x_i) \log_2 \left(p(y_j \mid x_i) \right)$$

$$H(Y \mid X) = P(x_1)H_1 + P(x_2)H_2 = 0.6(1.485) + 0.4(1.361)$$
$$H(Y \mid X) = 0.891 + 0.5444 = 1.4354$$

A transmitter (Tx) uses an alphabet consisting of five symbols: $A = \{a_1, a_2, a_3, a_4, a_5\}$. The receiver (Rx) uses an alphabet of four symbols: $B = \{b_1, b_2, b_3, b_4\}$. The joint probability matrix $P(A, B)$describing the probabilities of transmitting symbol $a_i$and receiving symbol $b_j$simultaneously is given by:

$$P(A,B) = \begin{bmatrix} 0.25 & 0 & 0 & 0 \\ 0.10 & 0.30 & 0 & 0 \\ 0 & 0.05 & 0.10 & 0 \\ 0 & 0 & 0.05 & 0.10 \\ 0 & 0.05 & 0 & 0 \end{bmatrix}$$

Using this joint probability matrix, compute the following information-theoretic measures: Source Entropy $H(A)$, Output Entropy $H(B)$, Joint Entropy $H(A, B)$, Conditional Entropy $H(B \mid A)$, Conditional Entropy $H(A \mid B)$ and Mutual Information $I(A; B)$.

Ans. Row = A, Column = B.

Check total probability:$0.25 + 0.10 + 0.30 + 0.05 + 0.10 + 0.05 + 0.05 + 0.10 = 1$

Valid probability matrix.

Compute input symbol Probabilities

$$P(a_1) = 0.25$$
$$P(a_2) = 0.10 + 0.30 = 0.40$$
$$P(a_3) = 0.05 + 0.10 = 0.15$$
$$P(a_4) = 0.05 + 0.10 = 0.15$$
$$P(a_5) = 0.05$$

Thus: $P(A) = \{0.25, 0.40, 0.15, 0.15, 0.05\}$

Compute output symbol Probabilities

$$P(b_1) = 0.25 + 0.10 + 0 + 0 + 0 = 0.35$$
$$P(b_2) = 0 + 0.30 + 0.05 + 0 + 0.05 = 0.40$$
$$P(b_3) = 0 + 0 + 0.10 + 0.05 + 0 = 0.15$$
$$P(b_4) = 0 + 0 + 0 + 0.10 + 0 = 0.10$$

Thus: $P(B) = \{0.35, 0.40, 0.15, 0.10\}$

Compute Entropies

Use: $H(X) = -\sum p(x) \log_2 p(x)$

Compute $H(A)$

$$H(A) = -(0.25 \log_2 0.25 + 0.40 \log_2 0.40 + 0.15 \log_2 0.15 + 0.15 \log_2 0.15 + 0.05 \log_2 0.05)$$
$$H(A) = 0.50 + 0.529 + 0.820 + 0.216 = 2.065 \text{ bits}$$

Compute $H(B)$

$$H(B) = -[0.35 \log_2 0.35 + 0.40 \log_2 0.40 + 0.15 \log_2 0.15 + 0.10 \log_2 0.10]$$

$$H(B) = 0.530 + 0.529 + 0.410 + 0.332 = 1.801 \text{ bits}$$

Compute Joint Entropy $\boldsymbol{H(A, B)}$

$$H(A, B) = -\sum_{i,j} P(a_i, b_j) \log_2 P(a_i, b_j)$$

$$H(A, B) = 0.25\log_2 0.25 + 0.10\log_2 0.10 + 0.30\log_2 0.30 + 0.15\log_2 0.05 + 0.20\log_2 0.10$$
$$H(A, B) = 0.50 + 0.332 + 0.521 + 0.648 + 0.664 = 2.665 \text{ bits}$$

Compute Conditional Entropies

$$H(B \mid A) = H(A, B) - H(A)$$
$$= 2.665 - 2.065 = 0.600 \text{ bits}$$

$$H(A \mid B) = H(A, B) - H(B)$$
$$= 2.665 - 1.801 = 0.864 \text{ bits}$$

Compute Mutual Information $\boldsymbol{I(A; B)}$

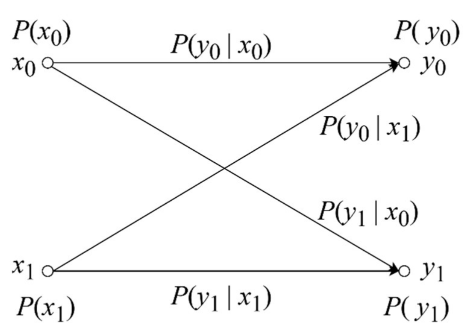
$$I(A; B) = H(A) - H(A \mid B)$$
$$I(A; B) = 2.065 - 0.864 = 1.201 \text{ bits}$$

2.10 Types of Channels:

- Binary Communication Channel
- Lossless Channel
- Deterministic Channel
- Noiseless Channel

2.11 Binary Communication Channel

A binary communication channel (shown in the below figure) is a digital communication system in which the transmitter sends information using only two possible symbols, typically represented as 0 and 1. These channels form the foundation of digital communications, computer networks, digital storage, and error-control coding.



Let the source is generate symbols x_0 (= 0) and x_1 (= 1) with probabilities $P(x_0)$ and $P(x_1)$, respectively. The channel output consists of two possible symbols y_0 and y_1, occurring with probabilities $P(y_0)$ and $P(y_1)$. If the input symbol probabilities and channel transition probabilities are known, the output probabilities are obtained as:

$$P(y_0) = P(y_0 \mid x_0)\, P(x_0) + P(y_0 \mid x_1)\, P(x_1)$$

$$P(y_1) = P(y_1 \mid x_0)\, P(x_0) + P(y_1 \mid x_1)\, P(x_1)$$

These relationships can be expressed compactly in matrix form as shown below:

$$\begin{bmatrix} P(y_0) \\ P(y_1) \end{bmatrix} = \begin{bmatrix} P(x_0) & P(x_1) \end{bmatrix} \begin{bmatrix} P(y_0 \mid x_0) & P(y_1 \mid x_0) \\ P(y_0 \mid x_1) & P(y_1 \mid x_1) \end{bmatrix}$$

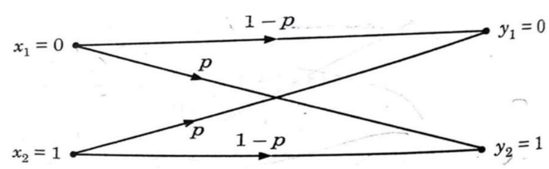
Binary communication channels are mainly classified as Binary Symmetric Channels (BSC) and Binary Erasure Channel (BEC).

2.11.1. Binary Symmetric Channel (BSC)

- A Binary Symmetric Channel (BSC) is one of the simplest and most widely used models in digital communication.
- It represents a channel that transmits binary symbols (0 and 1) and introduces errors with a fixed probability.
- A Binary Symmetric Channel is a discrete memoryless channel where: Input alphabet = {0, 1} and Output alphabet = {0, 1}.
- Each bit transmitted through the channel may be flipped (0→1 or 1→0) with probability p
- It is received correctly with probability $(1 - p)$, where p is called the crossover probability.

Example

$$P(Y \mid X) = \begin{bmatrix} 1-p & p \\ p & 1-p \end{bmatrix}$$

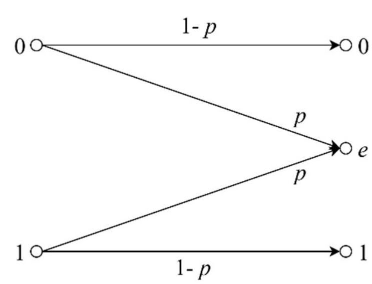


2.11.2. Binary Erasure Channel (BEC)

The analog of the binary symmetric channel in which some bits are lost (rather than corrupted) is the binary erasure channel. In this channel, a fraction p of the bits are erased. The receiver knows which bits have been erased. The binary erasure channel has two inputs and three outputs as shown in the below figure.

Example

$$P(Y \mid X) = \begin{bmatrix} 1-p & p & 0 \\ 0 & p & 1-p \end{bmatrix}$$



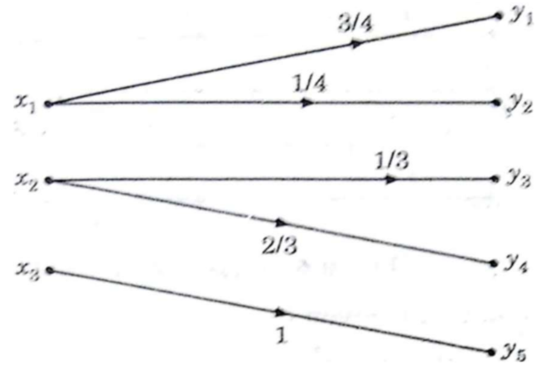
2.12 Lossless Channel:

- In lossless channel no source information is lost in transmission.

- Each output has received some information.
- Only one non-zero element in each column.

Example: Matrix example shown with non-zero entries and corresponding diagram.

$$P(Y \mid X) = \begin{bmatrix} \frac{3}{4} & \frac{1}{4} & 0 & 0 & 0 \\ 0 & 0 & \frac{1}{3} & \frac{2}{3} & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$



2.13 Deterministic Channel

- A Deterministic Channel is a special type of communication channel in information theory where each input symbol produces exactly one fixed output symbol with probability 1.
- There is no randomness, no noise, and no uncertainty in the output once the input is known.
- In Channel transition matrix, only one non-zero element is present in each row. This element must be unity.
- A channel is deterministic if:

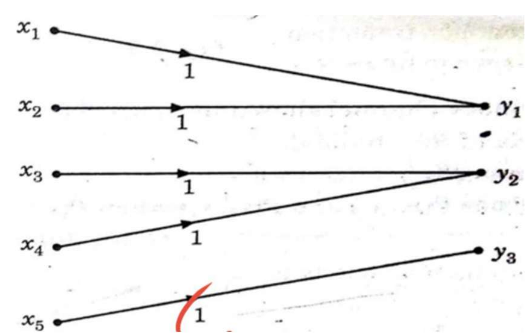
$$P(y_j \mid x_i) = \begin{cases} 1, & \text{for exactly one } y_j, \\ 0, & \text{for all others.} \end{cases}$$

This means:

- Each input symbol maps to one and only one output symbol.
- The mapping is perfectly predictable.
- The channel introduces no errors and no randomness.

Example

$$P(Y \mid X) = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$



2.14 Noiseless Channel:

A noiseless channel is a communication channel in which:

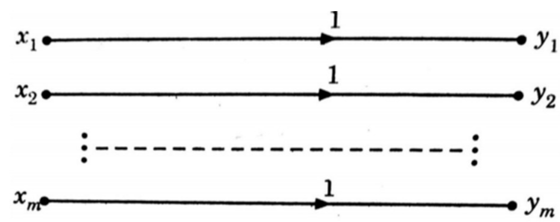
- The transmitted symbol is received correctly without any change.
- There is zero uncertainty in the output given the input.

- The channel introduces no distortion, no errors, and no randomness.
- A channel is noiseless, if it is both loss-less and deterministic.
- A channel is Noiseless if:

$$P(y_j \mid x_i) = \begin{cases} 1, & \text{if } y_j = \text{correct output for } x_i, \\ 0, & \text{for all others.} \end{cases}$$

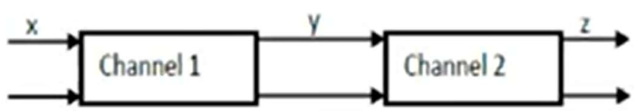
- Thus, $H(Y \mid X) = 0$.
- This means if you know what was transmitted, you know exactly what was received.
- Example

$$P(Y \mid X) = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$



2.15 Cascaded Channels

A cascaded channel refers to a communication system in which the transmitted signal passes through two or more channels sequentially, one after another as shown in below figure. Each channel introduces its own noise or distortion, and the overall channel behavior is determined by the combined effect of all channels in cascade. This model is widely used in digital communication, wireless systems, fading channels, and relay networks.



A signal X passes through Channel 1 and produces intermediate output Y. Channel 2: receives Y and produces final output Z. Then the communication chain is: $X \rightarrow Y \rightarrow Z$. The overall channel from X to Z is called the cascaded channel.

Each channel has its own transition probability matrix. Let:

$$P(y \mid x) = \text{matrix of Channel 1}$$

$$P(z \mid y) = \text{matrix of Channel 2}$$

Then the resultant channel from X to Z is:

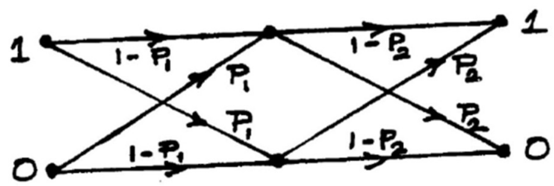
$$P(z \mid x) = \sum_{y} P(z \mid y) P(y \mid x)$$

This is a key formula in Information Theory, and it represents taking into account all possible intermediate paths.

Cascaded Binary Symmetric Channels (BSC):

A Binary Symmetric Channel has: Correct transmission probability: $(1 - p)$ and Error probability: p as shown in below figure. If two BSCs are cascaded with error probabilities p_1 and p_2, Then the overall error probability is:

$$p = p_1(1 - p_2) + p_2(1 - p_1)$$



This happens because:

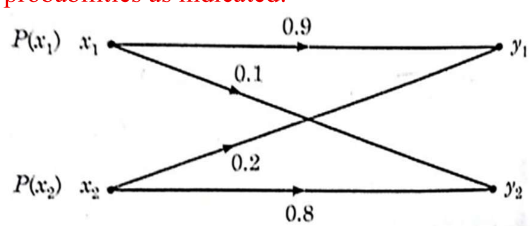
✓ Error may occur in channel 1

✓ Or error may occur in channel 2

✓ But two errors cancel each other (correct output)

The resulting channel is again a BSC, but with a different error probability.

Q1. Given the binary communication channel shown below, where the input symbols $x_1$and $x_2$are transmitted with transition probabilities as indicated:



Answer the following:

(a) Determine the channel matrix of the given binary channel.

(b) Compute the output probabilities $P(y_1)$and $P(y_2)$ when the input probabilities are

$$P(x_1) = P(x_2) = 0.5.$$

(c) Find the joint probabilities $P(x_1, y_2)$and $P(x_2, y_1)$for the same input probabilities.

Solution

(a) Channel Matrix

The channel matrix is defined as:

The channel transition probability matrix is:

$$P(Y \mid X) = \begin{bmatrix} P(y_1 \mid x_1) & P(y_2 \mid x_1) \\ P(y_1 \mid x_2) & P(y_2 \mid x_2) \end{bmatrix} = \begin{bmatrix} 0.9 & 0.1 \\ 0.2 & 0.8 \end{bmatrix}$$

(b) Find $P(y_1)$and $P(y_2)$

Using:

$$P(y_j) = \sum_i P(y_j \mid x_i) P(x_i)$$

Given $P(x_1) = P(x_2) = 0.5.$

Compute $P(y_1)$:

$$P(y_1) = 0.9(0.5) + 0.2(0.5)$$
$$P(y_1) = 0.45 + 0.10 = 0.55$$

Compute $P(y_2)$:

$$P(y_2) = 0.1(0.5) + 0.8(0.5)$$
$$P(y_2) = 0.05 + 0.40 = 0.45$$

Final output probabilities:

$$P(y_1) = 0.55, P(y_2) = 0.45$$

(c) Joint Probabilities $P(x_1, y_2)$and $P(x_2, y_1)$
Using:

$$P(x_i, y_j) = P(x_i)P(y_j \mid x_i)$$

1. Compute $P(x_1, y_2)$:

$$P(x_1, y_2) = P(x_1)P(y_2 \mid x_1)$$
$$P(x_1, y_2) = 0.5 \times 0.1 = 0.05$$

2. Compute $P(x_2, y_1)$:

$$P(x_2, y_1) = P(x_2)P(y_1 \mid x_2)$$
$$P(x_2, y_1) = 0.5 \times 0.2 = 0.10$$

Q2. A communication channel is described by the following channel transition matrix:

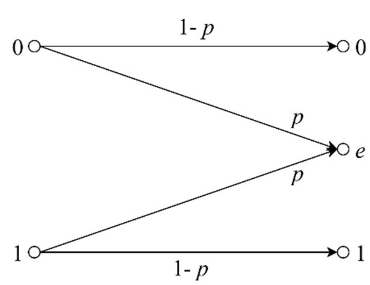
$$P(y \mid x) = \begin{bmatrix} 1-P & P & 0 \\ 0 & P & 1-P \end{bmatrix}$$

where the rows correspond to input symbols $x_1, x_2$and the columns correspond to output symbols y_1, y_2, y_3.

(a) Draw the channel diagram representing all possible transitions with their associated probabilities.
(b) If the source generates the two input symbols with equal probability, compute the output probabilities when $P = 0.2$.

Solution

(a) Channel Diagram



(b) Output Probabilities for $P = 0.2$
Given:

$$P(x_1) = P(x_2) = 0.5$$

Substitute $P = 0.2$.

1. Compute $P(y_1)$

$$P(y_1) = P(y_1 \mid x_1)P(x_1) + P(y_1 \mid x_2)P(x_2)$$
$$P(y_1) = (1 - P)(0.5) + (0)(0.5)$$
$$P(y_1) = (1 - 0.2)(0.5) = 0.8 \times 0.5 = 0.4$$

2. Compute $P(y_2)$

$$P(y_2) = P(y_2 \mid x_1)P(x_1) + P(y_2 \mid x_2)P(x_2)$$
$$P(y_2) = (0.2)(0.5) + (0.2)(0.5)$$